Environmental Product Declaration

EPD of multiple products, based on average product.
In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for, and conformity with ISO 21930:2017:

Steel for the reinforcement of concrete D500E and D500N

Diameter from 10-14 mm and 16-32 mm, listed on page 5

Jingye Steel Co., Ltd

Programme:	The International EPD® System, www.environdec.com
Programme operator:	EPD International AB
EPD registration number:	EPD-IES-0003562
Publication date:	2024-06-10
Valid until:	2029-06-09

An EPD should provide current information and may be updated if conditions change. The stated validity is therefore subject to the continued registration and publication at www.environdec.com

General information

Programme information

Programme:	The International EPD® System
Address:	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden
Website:	www.environdec.com
E-mail:	info@environdec.com

Accountabilities for PCR, LCA and independent, third-party verification
Product Category Rules (PCR)
CEN standard EN 15804 serves as the Core Product Category Rules (PCR)
Product Category Rules (PCR): PCR 2019:14 Construction products Version 1.3.3 [valid until: 2024-12-20]
PCR review was conducted by: The Technical Committee of the International EPD System. The review panel may be contacted via info@environdec.com.
Life Cycle Assessment (LCA)
LCA accountability: Sijia Yang and Rui Wang from IVL Swedish Environmental Research Institute
Third-party verification
Independent third-party verification of the declaration and data, according to ISO 14025:2006, via: ☒ EPD verification by individual verifier Third-party verifier: Pär Lindman and Sofia Lindroth, from Miljögiraff AB Approved by: The International EPD® System
Procedure for follow-up of data during EPD validity involves third party verifier: ☐ Yes ☒ No [Procedure for follow-up the validity of the EPD is at minimum required once a year with the aim of confirming whether the information in the EPD remains valid or if the EPD needs to be updated during its validity period. The follow-up can be organized entirely by the EPD owner or together with the original verifier via an agreement between the two parties. In both approaches, the EPD owner is responsible for the procedure being carried out. If a change that requires an update is identified, the EPD shall be re-verified by a verifier]

LCA practitioner:
IVL Swedish Environmental Research Institute Beijing Office
LCA operator: Sijia Yang (sijia.yang@ivl.se) and Rui Wang (rui.wang@ivl.se)
Contact: Juanjuan Yao (juanjuan.yao@ivl.se)

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but registered in different EPD programmes, or not compliant with EN 15804, may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison. For further information about comparability, see EN 15804 and ISO 14025.

It is discouraged to use of the results of modules A1-A3 without considering the results of module C.

Company information

Owner of the EPD:
Jingye Steel Co., Ltd
Pingshan town, Shijiazhuang City, Hebei Province, China

Contact:
Jinxiao Tan (+86 151 3212 3720)

Website:
https://www.hbjyjt.com/

Description of the organisation:
Jingye Group is a multinational group mainly engaged in the steel industry, Jingye Group owns Jingye Steel, British Steel, Ulanhot Steel, Guangdong Jingye Steel and Guangdong Yuebei steel. It is also engaged in steel deep processing, international trade, powder metallurgy 3D printing, tourism, hotels and other industries. It currently has 31 000 employees. In 2022, the group's sales revenue was 307.4 billion yuan, and the tax paid was 4.768 billion yuan. It ranked 112th in the top 500 enterprises nationwide and 46th of the top 500 manufacturing enterprises in China. It has topped the Fortune World 500 list for two consecutive years, ranking at 386th. In 2017, it was selected as one of the first batch of green factories by the Ministry of Industry and Information Technology, and was honored as "Meritorious Enterprise for the 40th Anniversary of the Reform and Opening up of the Iron and Steel Industry", "The Most Influential and Responsible Brand in Beijing-Tianjin-Hebei Region in 2017", "A+ Grade Competitive Iron and Steel Enterprise", and "A Grade Enterprise for Environmental Performance" .

Jingye Steel Co., Ltd. (Headquarters Steel) was established in 2002, affiliated to Jingye Group Co., Ltd., is the pillar enterprise of Jingye Group, is a high-tech enterprise, always put environmental protection, energy conservation and emission reduction in the first place, in 2017, was selected as the first batch of green factories by the Ministry of Industry and Information Technology, was named "2017 Beijing-Tianjin-Hebei Most Influential Responsible Brand", for many years won the "Hebei Famous Brand Product", "Consumer Trustworthy Product" and other titles, and has been continuously rated as a highly competitive enterprise of China's iron and steel enterprises by the Metallurgical Industry Planning and Research Institute. In the future, we will continue to adhere to green development and high-quality development, build China's high-quality construction steel base and circular economy demonstration base, and strive to develop into a modern iron and steel enterprise with advanced technology, excellent equipment, excellent quality, environmental protection standards and good benefits.

Product-related or management system-related certifications:
For products: China national standard “Steel for the reinforcement of concrete—Part 2:Hot rolled ribbed bars” (GB/T 1499.2-2018), Australian/New Zealand standard “Steel for the reinforcement of concrete” (ASNZS 4671-2019), British Standard “Steel for the reinforcement of concrete – Weldable reinforcing steel – Bar, coil and decoiled product – Specification" (BS 4449 2005+A3 2016), Korea standard “Reinforcing steel for reinforced concrete” (KSD 3504-2021).

For management system: Jingye Steel has certifications from the international quality management system, environmental management system (GB/T 24001-2016/ISO 14001:2015), occupational health and safety management system (GB/T 45001-2020/ISO 45001:2018), energy management system (GB/T 23331-2020/ISO 50001:2018 "energy management system requirements" and RB/T 103-2013 "Energy Management System Certification Requirements for Steel Enterprises"), ENTS carbon management system (T/CIECCPA 002-2021 "Carbon Management System Requirements and User Guide"). Energy Management System (GB/T 23331-2020/ISO 50001:2018 "Energy Management System Requirements" and RB/T 103-2013 "Energy Management System Certification Requirements

for Iron and Steel Enterprises"), ENTS Carbon Management System (T/CIECCPA 002-2021 "Carbon Management System Requirements and Usage Guidelines") and other certifications.

Name and location of production site:
Jingye Steel's Pingshan plant
Pingshan town, Shijiazhuang City, Hebei Province, China

Product information

Product name:
Steel for the reinforcement of concrete D500E and D500N (hereafter the rebar products).

Product identification:
In accordance with ASNZS 4671-2019.

Product description:
Jingye rebar products, with high welding, high strength and high toughness, are hot rolled and shaped, usually with a round cross-section and a smooth surface. It is the building material used to reinforce concrete structures. Reinforcing bars are embedded in the concrete to increase its tensile strength and provide structural integrity, and is one of the most widely used types of steel in civil engineering and construction. Reinforcing bars are available in different grades and sizes, each designed for specific applications and environments. The selection of the appropriate grade of hot rolled ribbed steel bar depends on factors such as the intended use of the concrete structure, exposure conditions and engineering requirements. With reference to the standards for the use of steel in construction and the life cycle of a building, the minimum service life of Jingye rebar is approximately 70 years.

This EPD is for multiple products, based on average product. The product included in this report covers the rebar products of D500E and D500N with the diameter from 10 to 40 mm. In this study, the average result is calculated from the average product, and the average product is obtained though dividing the total factory inputs or outputs by the total production of the factory.

The performance of the products in this family is shown in the table below, provided by Jingye, which is based on ASNZS 4671-2019.

Table 1 Product performance

Table 7.1(A) — Composition of reinforcing steels grades ≤ 500 MPa

Type of analysis	Chemical composition, % max.							
	All grades			Carbon equivalent value (*CEV*) for standard grades				
	C	P	S	250N	500L	500N	300E	500E
Cast analysis	0.22	0.050	0.050	0.43	0.39	0.44	0.43	0.49
Product analysis	0.24	0.055	0.055	0.45	0.41	0.46	0.45	0.51

Table 7.2(A) — Mechanical properties of reinforcing steel grades ≤ 500 MPa

Characteristic property		Specified value					Type of specified value
		250N[a]	500L[b]	500N	300E (Seismic)	500E (Seismic)	
Yield stress (MPa)	$R_{ek.L}$	≥ 250	≥ 500	≥ 500	≥ 300	≥ 500	C_{vL}: $P = 0.95$
	$R_{ek.U}$	—	≤ 750	≤ 650	≤ 380	≤ 600	C_{vU}: $P = 0.05$
Ratio	$(R_m/R_e)_{k.L}$	≥ 1.08	≥ 1.03	≥ 1.08	≥ 1.15	≥ 1.15	C_{vL}: $P = 0.90$
	$(R_m/R_e)_{k.U}$	—	—	—	≤ 1.50	≤ 1.40	C_{vU}: $P = 0.10$
Uniform elongation (%)	$A_{gt.k.L}$	≥ 5.0	≥ 1.5	≥ 5.0	≥ 15.0[c] ≥ 12.0[d]	≥ 10.0	C_{vL}: $P = 0.90$

[a] Grade 250N may be supplied as plain reinforcing steel in accordance with AS/NZS 3679.1.

[b] For 500L mesh (NZ only) and all 500L steels in coil form, the only requirement for $d < 5.0$ mm shall be $R_{ek.L} \geq 500$ MPa, except where smaller diameter bars are substituted as permitted by Table 7.6(A).

[c] Deformed reinforcing steel only.

[d] Plain reinforcing steel only.

(1) Please note that this table comes from Jingye's testing report, which is provided directly by Jingye. Since the inspection report is a PDF file, here is a screenshot of the file. Therefore, the table numbers in the screenshot do not exactly match the table numbers in this report. In order to avoid misunderstanding, a special explanation is provided here.

UN CPC code:
4124

Geographical scope:
Modules A1-A3 represent production of rebar products in Pingshan town, Shijiazhuang City, Hebei Province, China. Module A4 and A5 represent delivery from China to Oceania, mainly the New Zealand and Australia, and installation in these countries. Module B, C, and D represent the use and end-of-life treatment with benefits out of the system boundary of the products in Oceania countries.

LCA information

Declared unit:
One tonne of steel rebar.

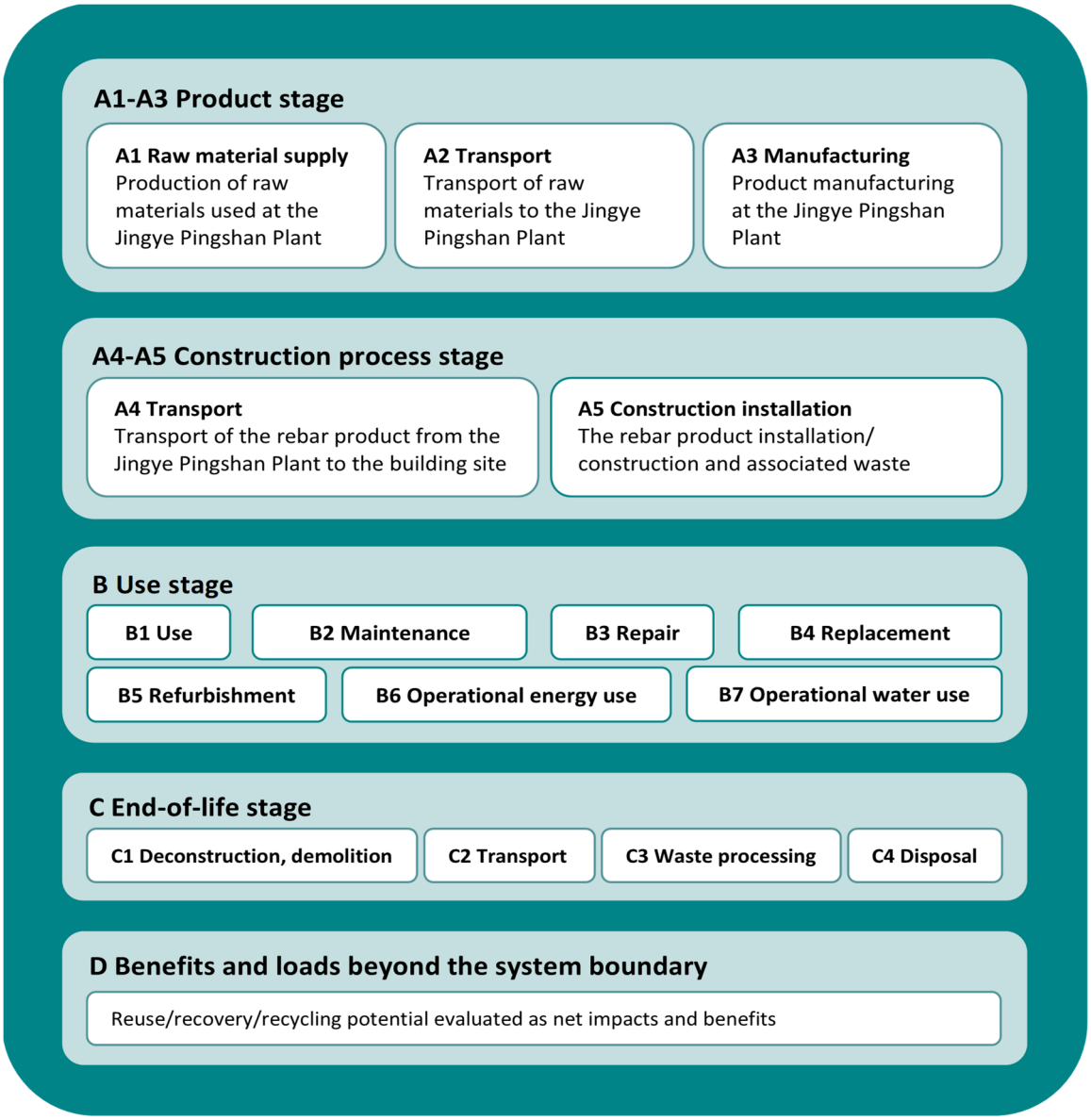
Reference service life:
70 years according to Jingye.

Time representativeness:
2023 (January to December).

Database(s) and LCA software used:
Managed LCA Content 2023.1 Databases (2023.1 Edition) and Ecoinvent 3.9.1, LCA for Expert (Gabi).

Description of system boundaries:
The scope of the EPD generated corresponds to “cradle to gate with options” which serves type (b) EPD, assessing the potential environmental impacts associated with the studied product. The information module included in the study is A1-A3, A4-A5, B1-B7, C1-C4, and D, no processes are omitted or excluded in this study.

System diagram:



Manufacturing processes:
As the manufacturing process is important to understand the whole studied system, the description of the main manufacturing processes for producing the studied product is explained as below. A flow chart of product manufacturing is shown below.

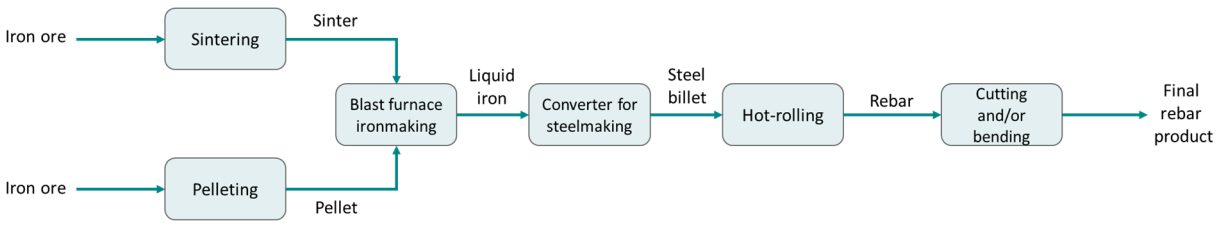


Figure 1 The main production process of studied product.

The rebar products are manufactured through the sintering, pelletising, blast furnace ironmaking, converter steelmaking, steel rolling, and cutting and/or bending (depending on needed) manufacturing processes. This flow chart shows the key inputs for each process. The basic raw material for the product is iron ore. The GWP-GHG of electricity mix for Hebei province in this study is 0.825 kg CO_2 eq./kWh.

More information:

Scenarios and additional technical information:

- The rebar products are manufactured in China, and the waste treatment of the waste generated during the manufacturing process was included in the system boundary.
- The rebar products are to and used in Oceania countries (mainly the New Zealand and Australia). For the A4 module, the specific data of the transportation information from the manufacturing plants to the overseas selling port is collected by Jingye, which is the specific data. From the selling port to the installation site, assumption is made due to the lack of data, which is 1 000 km by truck.
- Regarding A5 module, during the construction stage, the rebar products are required other materials to be installed together (e.g., concrete). However, due to the many uses of the rebar products (e.g., building a house or a bridge), the amount of other materials used is completely indeterminable and is different from case to case. Therefore, this LCA analysis is assumed that the consumption of additional materials in construction stage for installation is zero. Similarly, according to Jingye's information, the product could be installed by labour only and no energy is required to install it. Usually, such products may involve energy consumption during installation due to the use of other materials (e.g., concrete). In cases where this cannot be estimated, the consumption of energy during the construction phase is not considered in this study. Besides, according to the information provided by Jingye, the packaging of the rebar products is assumed to become to waste in this stage and is treated to end-of-life stage.
- In module B, the rebar products in this study do not generate emission and consume energy and water during the whole use phase, so the use phase environmental impact is deemed to be zero.
- The modelling of the C1 module is the hypothesis based on the A5 module, it is assumed that the consumption of additional materials and energy used in the deconstruction stage for installation is zero. The 15% of the rest of the waste rebar products would be sent to C4 for disposal, and 85% to recycling according to the worldsteel data. For the C2 module, conservative assumptions have been made that the waste product would be transported for 1 000 km by truck. For the waste processing C3 module, disposal C4 module, and D module (benefits and loads beyond the system boundary), the generic data has been applied. In detail, it is assumed the waste rebar products would be processed in C3 with a 3.01% mass loss.
- About module D calculation, since the rebar products have post-consumer material (only steel scrap) as the raw material, all steel scrap is excluded from the calculation of the benefit to avoid the double counting of the benefit. Meanwhile, this study does not consider the co-product allocation in this project, so there is no co-product allocation issue for the module D calculation. Besides, the mass loss during module D is assumed to be 5%, and we used the datasets and the plan directly from the Sphera database to calculate module D.

Allocation:
The rebar products have co-products generated throughout the production process, but they have low economic value. With the conservative consideration that co-products have low economic value, all impact is allocated to the main product in this LCA study and not allocated to the co-products, i.e., all environmental burdens are taken by the rebar product.

Cut-off rules:
The cut-off criteria established by the PCR is that data for elementary flows to and from the product system contributing to a minimum of 95% of the declared environmental impacts shall be included (not including processes that are explicitly outside the system boundary). No cut-off of elementary flows for the studied product has been applied in the study. From A5 and the following stages, some scenarios are set which may contain assumptions. Details could be found in the above section *"scenarios and additional technical information"*.

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results):

	Product stage			Construction process stage		Use stage							End of life stage				Resource recovery stage
	Raw material supply	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	**A1**	**A2**	**A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Modules declared	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Geography	CN	CN	CN	CN to OA	OA	OA	OA	OA	OA	OA	OA	OA	OA	OA	OA	OA	OA
Specific data used	>90%					-	-	-	-	-	-	-	-	-	-	-	-
Variation – products	<10%					-	-	-	-	-	-	-	-	-	-	-	-
Variation – sites	0%					-	-	-	-	-	-	-	-	-	-	-	-

Content information

This EPD is for multiple products, based on average product. The content declaration of the average product is listed below.

Average product:

Product components	Weight, kg	Post-consumer material, weight-% of total product	Biogenic material, kg C/product
Steel billet (hot-rolled)	1 000.00	10.38%	0.00
TOTAL	1 000.00	10.38%	0.00
Packaging materials	**Weight, kg**	**Weight-% (versus the product)**	**Weight biogenic carbon, kg C/product**
Strip (plastic)	1.20	0.00%	0.00
Steel wire	0.03	0.00%	0.00
TOTAL	1.23	0.00%	0.00

At the time of data collection, no substance included in the Candidate List of Substances of Very High Concern (SVHC) for authorization under the REACH Regulations is present in the products covered by this LCA and EPD either above the threshold for registration with the European Chemicals Agency or above 0.1% (wt/wt).

Depending on the PCR, in general, the production and end-of-life processes of infrastructure or capital goods used in the product system should be excluded, unless there is evidence that they are relevant in terms of their environmental impact, or when a generic LCI dataset includes infrastructure/capital goods, and it is not possible, within reasonable effort, to subtract the data on infrastructure/capital goods from this dataset (directly citation from section 4.3.2 of PCR 1.3.3). In this study, the infrastructure and capital goods are not included in the LCA analysis since they are used plenty of times for several years for the product manufacturing. According to the PCR, it should be excluded.

All results in this LCA analysis are calculated by the EN 15804+A2. The “EN 15804 reference package” is calculated based on EF 3.1.

Results of the environmental performance indicators

The estimated impact results are only relative statements which do not indicate the end points of the impact categories, exceeding threshold values, safety margins or risks.

Mandatory impact category indicators according to EN 15804

Results per functional or declared unit																
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-fossil	kg CO_2 eq.	2.16E+03	1.55E+02	3.09E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	7.49E+01	2.17E+00	2.64E+00	-7.61E+02
GWP-biogenic	kg CO_2 eq.	8.86E-01	5.30E-02	5.54E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.80E-02	9.81E-03	9.89E-03	4.36E+00
GWP-luluc	kg CO_2 eq.	2.95E-01	2.88E-03	5.23E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.01E-03	1.64E-02	8.19E-03	-9.77E-02
GWP-total	kg CO_2 eq.	2.16E+03	1.59E+02	3.09E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	7.82E+01	2.19E+00	2.65E+00	-7.57E+02
ODP	kg CFC 11 eq.	9.34E-07	9.87E-12	5.95E-13	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	6.60E-12	3.64E-12	6.70E-12	9.86E-10
AP	mol H^+ eq.	1.04E+02	1.39E+00	7.33E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.90E-01	1.13E-02	1.87E-02	-1.80E+00
EP-freshwater	kg P eq.	4.09E-02	2.16E-05	1.73E-07	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.27E-05	7.40E-06	5.31E-06	-1.71E-04
EP-marine	kg N eq.	3.18E+01	6.07E-01	2.72E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.38E-01	5.21E-03	4.83E-03	-2.88E-01
EP-terrestrial	mol N eq.	3.49E+02	6.66E+00	3.59E-03	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.52E+00	5.75E-02	5.32E-02	-2.59E+00
POCP	kg NMVOC eq.	8.51E+01	1.52E+00	6.33E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.76E-01	1.41E-02	1.46E-02	-1.17E+00
ADP-minerals& metals*	kg Sb eq.	7.51E-04	7.62E-07	5.57E-09	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.83E-07	2.33E-06	1.22E-07	-4.15E-03
ADP-fossil*	MJ	2.15E+04	2.11E+03	2.65E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.06E+03	4.28E+01	3.51E+01	-7.30E+03
WDP*	m^3	8.51E+01	5.60E-01	2.85E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.12E-01	4.23E-01	2.90E-01	-4.95E+01
Acronyms	GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption															

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.*

Additional mandatory and voluntary impact category indicators

Results per functional or declared unit																
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-GHG[1]	kg CO_2 eq.	2.12E+03	1.53E+02	3.08E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	7.38E+01	2.14E+00	2.60E+00	-7.31E+02

According to the PCR and EN 15804, the primary energy use indicators result needs to be balanced for the construction product. Since the studied product only have the plastic package related to this issue, the result of “Use of non-renewable primary energy (PENRE)” and “Non-renewable primary energy resources used as raw materials (PENRM)”, according to the option A of the Annex 3 of the PCR.

Resource use indicators

Results per functional or declared unit																
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
PERE	MJ	8.51E+02	7.36E+00	3.27E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.84E+00	3.98E+00	5.72E+00	2.88E+02
PERM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	8.51E+02	7.36E+00	3.27E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.84E+00	3.98E+00	5.72E+00	2.88E+02
PENRE	MJ	2.15E+04	2.12E+03	5.36E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.06E+03	4.29E+01	3.51E+01	-7.30E+03
PENRM	MJ	5.10E+01	0.00E+00	-5.10E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ	2.15E+04	2.12E+03	2.65E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.06E+03	4.29E+01	3.51E+01	-7.30E+03
SM	kg	4.56E+02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m^3	2.08E+00	1.31E-02	6.78E-03	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	6.20E-03	1.22E-02	8.87E-03	-7.42E+01
Acronyms	PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water															

[1] This indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic CO_2 is set to zero.

Waste indicators

Results per functional or declared unit																
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Hazardous waste disposed	kg	5.14E-04	1.13E-09	1.17E-11	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	7.66E-10	-1.11E-10	7.65E-10	-5.46E-05
Non-hazardous waste disposed	kg	1.48E+01	6.18E-02	3.02E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.29E-02	1.13E-02	1.76E+02	8.83E+01
Radioactive waste disposed	kg	4.42E-02	4.71E-04	4.81E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.54E-04	5.75E-04	4.00E-04	8.00E-04

Output flow indicators

Results per functional or declared unit																
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Components for re-use	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Material for recycling	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Materials for energy recovery	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy, electricity	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy, thermal	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

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